

Problem Set 1 Solutions

Exercise 3

1. Solve in $\mathbb{R}$:

1. $x - 1 \leq \sqrt{x + 2}$

2. $(2x - 7) \ln(x + 1) > 0$

3. $\ln\left(\frac{x+1}{3x-5}\right) \leq 0$

4. $x + \frac{1}{x} \geq 2$

. 1. Domain: $x \geq -2$. For $x \in [-2, 1)$, inequality holds. For $x \geq 1$:

$$(x - 1)^2 \leq x + 2 \implies x^2 - 3x - 1 \leq 0 \implies x \in \left[1, \frac{3 + \sqrt{13}}{2}\right]$$

Solution: $\left[-2, \frac{3 + \sqrt{13}}{2}\right]$

2. Domain: $x > -1$. Solution:

$$x \in (-1, 0) \cup (3.5, \infty)$$

3. Domain: $x < -1$ or $x > \frac{5}{3}$. Solution:

$$\frac{x + 1}{3x - 5} \leq 1 \implies x \in (-\infty, -1) \cup [3, \infty)$$

4. Domain: $x \neq 0$. For $x > 0$:

$$x + \frac{1}{x} \geq 2 \iff (x - 1)^2 \geq 0$$

Solution: $(0, \infty)$

Exercise 4

2. Prove that for any $x > 0$:

$$x - \frac{x^3}{6} \leq \ln(1 + x) \leq x$$

. For the upper bound:

$$f(x) = \ln(1 + x) - x \implies f'(x) = \frac{1}{1 + x} - 1 \leq 0 \quad \text{for } x > 0$$

Since $f(0) = 0$, $f(x) \leq 0$ for $x > 0$.

For the lower bound:

$$g(x) = \ln(1 + x) - x + \frac{x^3}{6} \implies g'(x) = \frac{1}{1 + x} - 1 + \frac{x^2}{2} \geq 0 \quad \text{for } x > 0$$

Since $g(0) = 0$, $g(x) \geq 0$ for $x > 0$.

Exercise 5

3. Let a, b, c be real positive numbers. Show that:

1. $a^2 + b^2 + c^2 \geq ab + bc + ca$

2. $(a + b + c)^2 \geq 3(ab + bc + ca)$

. 1. **Proof of $a^2 + b^2 + c^2 \geq ab + bc + ca$:**

Starting from the fact that squares are always non-negative:

$$(a - b)^2 \geq 0 \Rightarrow a^2 - 2ab + b^2 \geq 0$$

$$(b - c)^2 \geq 0 \Rightarrow b^2 - 2bc + c^2 \geq 0$$

$$(c - a)^2 \geq 0 \Rightarrow c^2 - 2ca + a^2 \geq 0$$

Adding these three inequalities:

$$2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca \geq 0$$

Dividing by 2 gives the desired result:

$$a^2 + b^2 + c^2 \geq ab + bc + ca$$

2. **Proof of $(a + b + c)^2 \geq 3(ab + bc + ca)$:**

First expand the left side:

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

From part (a), we know:

$$a^2 + b^2 + c^2 \geq ab + bc + ca$$

Adding $2(ab + bc + ca)$ to both sides:

$$a^2 + b^2 + c^2 + 2(ab + bc + ca) \geq 3(ab + bc + ca)$$

Substituting the expanded form:

$$(a + b + c)^2 \geq 3(ab + bc + ca)$$

Exercise 6

4. For non-negative real numbers a, b , prove:

$$8(a^4 + b^4) \geq (a + b)^4$$

Proof. Let $p = a + b$ and $q = a - b$. Then:

$$a^4 + b^4 = \left(\frac{p+q}{2}\right)^4 + \left(\frac{p-q}{2}\right)^4 = \frac{p^4 + 6p^2q^2 + q^4}{8}$$

Substituting into the original inequality:

$$\begin{aligned} 8 \left(\frac{p^4 + 6p^2q^2 + q^4}{8} \right) &\geq p^4 \\ \Rightarrow p^4 + 6p^2q^2 + q^4 &\geq p^4 \\ \Rightarrow q^2(6p^2 + q^2) &\geq 0 \end{aligned}$$

The inequality holds. □

Exercise 7

5. Let $a_1, \dots, a_n$ be real numbers with $a_k > -1$ for all k , and either all $a_k \geq 0$ or all $a_k \leq 0$. Prove:

$$\prod_{k=1}^n (1 + a_k) \geq 1 + \sum_{k=1}^n a_k$$

Proof. We proceed by mathematical induction on n .

Base case ($n = 1$):

$$1 + a_1 \geq 1 + a_1$$

holds trivially.

Inductive step: Assume the inequality holds for n . Then for $n + 1$:

$$\prod_{k=1}^{n+1} (1 + a_k) = (1 + a_{n+1}) \prod_{k=1}^n (1 + a_k) \geq (1 + a_{n+1}) \left(1 + \sum_{k=1}^n a_k \right)$$

Expanding the right side:

$$1 + \sum_{k=1}^n a_k + a_{n+1} + a_{n+1} \sum_{k=1}^n a_k \geq 1 + \sum_{k=1}^{n+1} a_k$$

since $a_{n+1} \sum_{k=1}^n a_k \geq 0$ (as all a_k have the same sign).

Equality holds when either:

- All $a_k = 0$, or
- Exactly one $a_k \neq 0$ and the rest are zero.

□

6 (Bernoulli's Inequality). Deduce that for any $a > -1$ and $n \in \mathbb{N}$:

$$(1 + a)^n \geq 1 + na$$

Proof. Take $a_1 = \dots = a_n = a$ in the previous result. The condition on the signs is satisfied since either:

- $a \geq 0$ (all non-negative), or
- $-1 < a \leq 0$ (all non-positive)

The inequality becomes:

$$(1 + a)^n \geq 1 + na$$

with equality when $a = 0$ or $n = 1$. □

Exercise 8

7. For positive real numbers $x_1, \dots, x_n$, prove:

$$\frac{1}{n} \sum_{i=1}^n x_i \geq \left(\prod_{i=1}^n x_i \right)^{1/n} \geq \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

with equality if and only if all x_i are equal.

Proof. We prove the AM-GM inequality by forward-backward induction.

Definitions:

$$A_n = \frac{1}{n} \sum_{k=1}^n x_k \quad (\text{Arithmetic mean})$$

$$G_n = \left(\prod_{k=1}^n x_k \right)^{1/n} \quad (\text{Geometric mean})$$

Base Case (n=1): Trivially true since $A_1 = x_1 = G_1$.

Base Case (n=2): For $x_1, x_2 > 0$:

$$0 \leq (\sqrt{x_1} - \sqrt{x_2})^2 = x_1 - 2\sqrt{x_1 x_2} + x_2$$

which rearranges to:

$$\frac{x_1 + x_2}{2} \geq \sqrt{x_1 x_2}$$

Forward Induction: Assume true for $n = 2^k$. For $n = 2^{k+1} = 2m$ where $m = 2^k$:

$$\begin{aligned} \left(\prod_{i=1}^{2m} x_i \right)^{1/2m} &= \left(\left(\prod_{i=1}^m x_i \right)^{1/m} \left(\prod_{i=m+1}^{2m} x_i \right)^{1/m} \right)^{1/2} \\ &\leq \frac{1}{2} \left(\frac{\sum_{i=1}^m x_i}{m} + \frac{\sum_{i=m+1}^{2m} x_i}{m} \right) \\ &= \frac{\sum_{i=1}^{2m} x_i}{2m} \end{aligned}$$

Backward Induction: Assume true for $n = k$. For $n = k - 1$, set $x_k = G_{k-1}$:

$$\left(\prod_{i=1}^k x_i \right)^{1/k} \leq \frac{1}{k} \left(\sum_{i=1}^{k-1} x_i + G_{k-1} \right)$$

Substituting $G_{k-1}^{k-1} = \prod_{i=1}^{k-1} x_i$:

$$G_{k-1} \leq \frac{(k-1)A_{k-1} + G_{k-1}}{k} \implies G_{k-1} \leq A_{k-1}$$

HM-GM Inequality: Applying AM-GM to reciprocals $y_i = 1/x_i$:

$$\frac{1}{n} \sum_{i=1}^n y_i \geq \left(\prod_{i=1}^n y_i \right)^{1/n} \implies \frac{n}{\sum_{i=1}^n \frac{1}{x_i}} \leq \left(\prod_{i=1}^n x_i \right)^{1/n}$$

Conclusion: By induction, for all $n \geq 1$:

$$\boxed{\frac{1}{n} \sum_{i=1}^n x_i \geq \left(\prod_{i=1}^n x_i \right)^{1/n} \geq \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}}$$

with equality if and only if all x_i are equal. □

Exercise 9

8. Let $a_1 \geq a_2 \geq \dots \geq a_n$ and $b_1 \geq b_2 \geq \dots \geq b_n$ be sequences of real numbers. Prove:

$$\frac{1}{n} \sum_{k=1}^n a_k b_k \geq \left(\frac{1}{n} \sum_{k=1}^n a_k \right) \left(\frac{1}{n} \sum_{k=1}^n b_k \right)$$

Proof. Consider the following expansion for any $j, k \in \{1, 2, \dots, n\}$:

$$(a_j - a_k)(b_j - b_k) \geq 0$$

This inequality holds because:

- If $j \leq k$, then $a_j \geq a_k$ and $b_j \geq b_k$, so both factors are non-negative
- If $j > k$, then $a_j \leq a_k$ and $b_j \leq b_k$, so both factors are non-positive

Summing over all pairs (j, k) :

$$\sum_{j=1}^n \sum_{k=1}^n (a_j - a_k)(b_j - b_k) \geq 0$$

Expanding the product:

$$\sum_{j=1}^n \sum_{k=1}^n (a_j b_j - a_j b_k - a_k b_j + a_k b_k) \geq 0$$

Simplifying the sums:

$$2n \sum_{k=1}^n a_k b_k - 2 \left(\sum_{k=1}^n a_k \right) \left(\sum_{k=1}^n b_k \right) \geq 0$$

Dividing by $2n^2$:

$$\frac{1}{n} \sum_{k=1}^n a_k b_k - \left(\frac{1}{n} \sum_{k=1}^n a_k \right) \left(\frac{1}{n} \sum_{k=1}^n b_k \right) \geq 0$$

Thus we obtain the desired inequality:

$$\frac{1}{n} \sum_{k=1}^n a_k b_k \geq \left(\frac{1}{n} \sum_{k=1}^n a_k \right) \left(\frac{1}{n} \sum_{k=1}^n b_k \right)$$

□

Equality condition: Equality holds if and only if either:

- All a_k are equal, or
- All b_k are equal.

Exercise 10

9. Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be real numbers. Prove:

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right)$$

Proof. Consider the following quadratic expression in λ :

$$P(\lambda) = \sum_{i=1}^n (a_i \lambda + b_i)^2 \geq 0$$

Expanding the square:

$$P(\lambda) = \left(\sum_{i=1}^n a_i^2 \right) \lambda^2 + 2 \left(\sum_{i=1}^n a_i b_i \right) \lambda + \sum_{i=1}^n b_i^2 \geq 0$$

Since $P(\lambda)$ is always non-negative, its discriminant must satisfy:

$$\Delta = 4 \left(\sum_{i=1}^n a_i b_i \right)^2 - 4 \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right) \leq 0$$

Dividing by 4 and rearranging yields:

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right)$$

Equality condition: Holds if and only if the sequences are linearly dependent, i.e., there exists α, β (not both zero) such that $\alpha a_i + \beta b_i = 0$ for all i . $\square$

Exercise 11

10. Let $x_1, x_2, \dots, x_n$ be real numbers satisfying:

$$\sum_{i=1}^n x_i = n \quad \text{and} \quad \sum_{i=1}^n x_i^2 = n.$$

Prove that $x_1 = x_2 = \dots = x_n = 1$.

Proof. Consider the sum of squared deviations from the mean:

$$\sum_{i=1}^n (x_i - 1)^2 = \sum_{i=1}^n (x_i^2 - 2x_i + 1) = \left(\sum_{i=1}^n x_i^2\right) - 2\left(\sum_{i=1}^n x_i\right) + n$$

Substituting the given conditions:

$$= n - 2n + n = 0$$

Since each term $(x_i - 1)^2$ is non-negative, their sum being zero implies:

$$(x_i - 1)^2 = 0 \quad \text{for all } i = 1, \dots, n$$

Thus:

$$x_i = 1 \quad \text{for all } i = 1, \dots, n$$

$\square$

Exercise 12

11. Let $x_1, \dots, x_n \in [0, 1]$ with $S = \sum_{i=1}^n x_i < 1$. Prove:

$$1 + S \leq \prod_{i=1}^n (1 + x_i) \leq \frac{1}{1 - S}$$

Proof. Part 1: Proof of the lower bound $1 + S \leq \prod_{i=1}^n (1 + x_i)$

We proceed by induction on n .

Base case (n=1): For $x_1 \in [0, 1]$, we have:

$$1 + x_1 \leq 1 + x_1$$

which holds trivially.

Inductive step: Assume the inequality holds for $n = k$, i.e.:

$$\prod_{i=1}^k (1 + x_i) \geq 1 + \sum_{i=1}^k x_i$$

For $n = k + 1$:

$$\prod_{i=1}^{k+1} (1 + x_i) = (1 + x_{k+1}) \prod_{i=1}^k (1 + x_i) \geq (1 + x_{k+1}) \left(1 + \sum_{i=1}^k x_i \right)$$

Expanding the right side:

$$= 1 + \sum_{i=1}^k x_i + x_{k+1} + x_{k+1} \sum_{i=1}^k x_i \geq 1 + \sum_{i=1}^{k+1} x_i$$

since $x_{k+1} \sum_{i=1}^k x_i \geq 0$.

Part 2: Proof of the upper bound $\prod_{i=1}^n (1 + x_i) \leq \frac{1}{1-S}$

Again by induction on n .

Base case (n=1): For $x_1 \in [0, 1)$:

$$1 + x_1 \leq \frac{1}{1 - x_1} \iff (1 - x_1)(1 + x_1) = 1 - x_1^2 \leq 1$$

which holds since $x_1^2 \geq 0$.

Inductive step: Assume the inequality holds for $n = k$. For $n = k + 1$:

$$\prod_{i=1}^{k+1} (1 + x_i) \leq \frac{1 + x_{k+1}}{1 - \sum_{i=1}^k x_i}$$

We need to show:

$$\frac{1 + x_{k+1}}{1 - S_k} \leq \frac{1}{1 - S_{k+1}}$$

where $S_k = \sum_{i=1}^k x_i$ and $S_{k+1} = S_k + x_{k+1}$. Cross-multiplying (valid since denominators are positive):

$$(1 + x_{k+1})(1 - S_{k+1}) \leq 1 - S_k$$

Expanding and simplifying:

$$1 - S_k - x_{k+1}S_k + x_{k+1} - x_{k+1}^2 \leq 1 - S_k \iff x_{k+1}(1 - S_k - x_{k+1}) \leq 0$$

which holds because $1 - S_k - x_{k+1} \geq 1 - S_{k+1} > 0$ and $x_{k+1} \geq 0$.

□

$$\boxed{1 + \sum_{i=1}^n x_i \leq \prod_{i=1}^n (1 + x_i) \leq \frac{1}{1 - \sum_{i=1}^n x_i}}$$

Exercise 13

12. Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be real numbers. Prove:

$$\left(\sum_{k=1}^n (a_k + b_k)^2 \right)^{1/2} \leq \left(\sum_{k=1}^n a_k^2 \right)^{1/2} + \left(\sum_{k=1}^n b_k^2 \right)^{1/2}$$

Proof. We begin by expanding the square:

$$\sum_{k=1}^n (a_k + b_k)^2 = \sum_{k=1}^n (a_k^2 + 2a_k b_k + b_k^2) = \sum_{k=1}^n a_k^2 + 2 \sum_{k=1}^n a_k b_k + \sum_{k=1}^n b_k^2$$

By the Cauchy-Schwarz inequality:

$$\sum_{k=1}^n a_k b_k \leq \left(\sum_{k=1}^n a_k^2 \right)^{1/2} \left(\sum_{k=1}^n b_k^2 \right)^{1/2}$$

Substituting this bound:

$$\sum_{k=1}^n (a_k + b_k)^2 \leq \sum_{k=1}^n a_k^2 + 2 \left(\sum_{k=1}^n a_k^2 \right)^{1/2} \left(\sum_{k=1}^n b_k^2 \right)^{1/2} + \sum_{k=1}^n b_k^2$$

This can be rewritten as a perfect square:

$$\left(\left(\sum_{k=1}^n a_k^2 \right)^{1/2} + \left(\sum_{k=1}^n b_k^2 \right)^{1/2} \right)^2$$

Taking square roots of both sides (which preserves the inequality since all terms are non-negative) yields:

$$\left(\sum_{k=1}^n (a_k + b_k)^2 \right)^{1/2} \leq \left(\sum_{k=1}^n a_k^2 \right)^{1/2} + \left(\sum_{k=1}^n b_k^2 \right)^{1/2}$$

□

Exercise 14

13. Let $a, b > 0$ and $n \in \mathbb{N}$. Prove:

$$(a + b)^n \geq a^n + na^{n-1}b$$

Proof. By the Binomial Theorem, we have the expansion:

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

Explicitly writing the first two terms and the remainder:

$$= a^n + na^{n-1}b + \sum_{k=2}^n \binom{n}{k} a^{n-k} b^k$$

Since $a, b > 0$ and binomial coefficients satisfy $\binom{n}{k} \geq 0$ for all k , the remaining terms are non-negative:

$$\sum_{k=2}^n \binom{n}{k} a^{n-k} b^k \geq 0$$

Therefore:

$$(a + b)^n \geq a^n + na^{n-1}b$$

□